Decision-dependent distributionally robust optimization for resilient and sustainable multimodal transportation under endogenous uncertainty

1 Allocation of flexible budget data

Table 1: Results with 10 orders under alternative budget allocations

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
10	1000	576977	1	0	70	0	0.8328%
10	2000	555962	2	0	0	0	0.9322%
10	3000	551286	1	6	0	0	0.7560%
10	4000	549842	2	4	0	0	0.8400%
10	5000	546355	1	7	1	23	0.9961%
10	6000	549822	1	8	0	23	0.9730%
10	7000	547623	1	8	0	23	0.5564%
10	8000	547611	1	7	0	23	0.4306%
10	9000	549543	1	7	14	23	0.9766%
10	10000	546350	1	7	1	23	0.9768%
10	11000	546360	1	7	1	23	0.4822%
10	12000	548973	1	7	0	35	0.6948%
10	13000	546606	1	7	0	23	0.3920%
10	14000	549237	1	6	0	0	0.9202%
10	15000	549237	1	6	0	0	0.9202%
10	16000	548297	1	8	1	53	0.8679%
10	17000	546728	1	7	0	23	0.9522%
10	18000	550298	0	11	3	88	0.9185%
10	19000	549177	1	7	2	23	0.9076%
10	20000	549177	1	7	2	23	0.9076%
10	21000	549177	1	7	2	23	0.9076%
10	22000	549177	1	7	2	23	0.9076%
10	23000	549177	1	7	2	23	0.9076%
10	24000	549177	1	7	2	23	0.9076%
10	25000	546349	1	7	1	23	0.9821%
10	26000	549541	1	7	14	23	0.9725%
10	27000	549541	1	7	14	23	0.9725%
10	28000	549541	1	7	14	23	0.9725%
10	29000	548973	1	7	0	35	0.6858%
10	30000	549204	1	7	1	23	0.9999%
10	31000	549204	1	7	1	23	0.9998%
10	32000	549204	1	7	1	23	0.9981%
10	33000	549204	1	7	1	23	0.9981%
10	34000	549204	1	7	1	23	0.9981%

Table 1: Results with 10 orders under alternative budget allocations (continued)

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
10	35000	549117	1	7	2	23	0.9171%
10	36000	549117	1	7	2	23	0.9171%
10	37000	549117	1	7	2	23	0.9171%
10	38000	547777	1	9	3	63	0.7686%
10	39000	547777	1	9	3	63	0.7686%
10	40000	546354	1	7	1	23	0.9878%
10	41000	549520	1	8	2	33	0.9974%
10	42000	549520	1	8	2	33	0.9974%
10	43000	549520	1	8	2	33	0.9974%
10	44000	549520	1	8	2	33	0.9974%
10	45000	549520	1	8	2	33	0.9974%
10	46000	549520	1	8	2	33	0.9974%
10	47000	549520	1	8	2	33	0.9974%
10	48000	549520	1	8	2	33	0.9974%
10	49000	549520	1	8	2	33	0.9974%
10	50000	549520	1	8	2	33	0.9974%
10	55000	547349	1	7	1	23	0.6930%
10	60000	549236	1	7	1	23	0.9200%
10	65000	549236	1	7	1	23	0.9200%
10	70000	549236	1	7	1	23	0.9200%
10	75000	548382	1	7	1	23	0.9991%
10	80000	549236	1	7	1	23	0.9200%
10	85000	546349	1	7	1	23	0.9996%
10	90000	546349	1	7	1	23	0.9996%
10	95000	546349	1	7	1	23	0.9970%
10	100000	546353	1	7	1	23	0.5114%

Table 2: Results with 20 orders under alternative budget allocations

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
20	1000	1088168	1	0	153	0	0.8434%
20	2000	1055682	2	0	69	0	0.9584%
20	3000	1031185	1	7	28	2	0.9052%
20	4000	1027056	2	7	0	0	0.8648%
20	5000	1023980	3	7	0	0	0.9474%
20	6000	1020051	4	7	0	0	0.6623%
20	7000	1022478	5	7	0	0	0.8811%
20	8000	1022134	6	7	48	2	0.9484%
20	9000	1020247	5	9	1	111	0.9997%
20	10000	1022750	6	5	0	0	0.9486%
20	11000	1019741	6	5	0	0	0.7168%
20	12000	1022577	6	6	0	0	0.9894%
20	13000	1015771	7	4	0	0	0.3045%
20	14000	1019672	6	5	0	0	0.6904%
20	15000	1021457	7	7	0	99	0.9026%
20	16000	1022072	5	5	0	0	0.9385%
20	17000	1018710	6	7	27	2	0.6268%
20	18000	1018810	7	3	0	0	0.6273%

Table 2: Results with 20 orders under alternative budget allocations (continued)

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
20	19000	1016765	7	6	0	0	0.4274%
20	20000	1017873	6	3	0	0	0.5233%
20	21000	1015799	7	3	0	0	0.3293%
20	22000	1021951	6	5	0	0	0.9120%
20	23000	1021038	6	6	0	66	0.8479%
20	24000	1021136	6	3	0	0	0.8018%
20	25000	1021136	6	3	0	0	0.8018%
20	26000	1018042	6	4	0	0	0.5012%
20	27000	1021094	6	6	0	0	0.7910%
20	28000	1021617	7	5	0	33	0.8675%
20	29000	1020589	6	7	26	2	0.8025%
20	30000	1021567	6	4	0	0	0.8898%
20	31000	1015704	6	7	0	2	0.3246%
20	32000	1021123	5	6	0	0	0.8504%
20	33000	1021123	5	6	0	0	0.8504%
20	34000	1022387	6	6	0	0	0.9254%
20	35000	1022387	6	6	0	0	0.9254%
20	36000	1022403	6	4	0	0	0.9640%
20	37000	1018807	7	5	0	0	0.6094%
20	38000	1017504	6	3	0	0	0.4550%
20	39000	1020687	6	4	0	0	0.7867%
20	40000	1021693	6	6	0	0	0.8852%
20	41000	1021693	6	6	0	0	0.8852%
20	42000	1021682	6	7	41	2	0.9237%
20	43000	1022685	6	8	2	49	0.9994%
20	44000	1016727	7	4	0	0	0.4197%
20	45000	1015454	6	7	2	2	0.2769%
20	46000	1017897	7	4	0	0	0.5082%
20	47000	1016731	7	4	0	0	0.3974%
20	48000	1022325	6	5	0	0	0.9485%
20	49000	1016542	7	5	0	0	0.3198%
20	50000	1015776	7	4	0	0	0.3273%
20	55000	1017238	6	4	0	0	0.4453%
20	60000	1017986	6	7	2	9	0.8501%
20	65000	1016845	6	6	0	0	0.4387%
20	70000	1019943	6	7	1	33	0.7070%
20	75000	1018570	6	7	27	2	0.6113%
20	80000	1020301	6	7	39	2	0.7781%
20	85000	1018034	7	5	1	33	0.5474%
20	90000	1020257	7	3	0	0	0.6994%
20	95000	1019922	6	5	0	0	0.7284%
20	100000	1022597	6	8	2	49	0.9999%

Table 3: Results with 30 orders under alternative budget allocations

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
30	1000	1510079	1	0	147	0	0.5393%
30	2000	1467133	2	0	95	0	0.9762%

Table 3: Results with 30 orders under alternative budget allocations (continued)

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
30	3000	1437205	1	9	22	6	0.9876%
30	4000	1427982	2	9	6	0	0.8804%
30	5000	1420920	3	9	14	6	0.6999%
30	6000	1417346	4	9	6	0	0.5379%
30	7000	1413663	5	9	2	6	0.4839%
30	8000	1421411	6	10	27	6	0.9850%
30	9000	1419987	7	8	14	0	0.9329%
30	10000	1420074	7	8	20	2	0.9545%
30	11000	1419513	8	7	6	0	0.8900%
30	12000	1416902	9	6	6	0	0.7210%
30	13000	1417757	9	7	6	0	0.7533%
30	14000	1416673	10	7	3	74	0.7041%
30	15000	1418887	9	9	17	115	0.8305%
30	16000	1420079	8	6	52	16	0.9999%
30	17000	1413117	8	7	0	4	0.4616%
30	18000	1417035	9	9	10	101	0.7677%
30	19000	1417053	8	8	12	15	0.9807%
30	20000	1419316	9	7	35	0	0.8956%
30	21000	1417592	7	8	45	2	0.8636%
30	22000	1420565	6	11	14	79	0.9999%
30	23000	1413865	8	7	15	2	0.5697%
30	24000	1417243	8	7	13	10	0.9947%
30	25000	1413277	8	6	21	0	0.5200%
30	26000	1418626	8	6	38	4	0.8834%
30	27000	1411671	8	7	15	0	0.3940%
30	28000	1416589	8	7	32	2	0.7447%
30	29000	1418875	8	7	6	0	0.8550%
30	30000	1415475	8	7	6	0	0.6083%
30	31000	1414938	8	6	6	0	0.5716%
30	32000	1415992	8	8	17	95	0.6935%
30	33000	1412911	8	7	18	0	0.5378%
30	34000	1413683	8	8	17	0	0.5504%
30	35000	1415886	10	7	9	6	0.6626%
30	36000	1415166	9	8	7	2	0.6586%
30	37000	1411490	8	7	13	4	0.3917%
30	38000	1416584	8	7	26	2	0.7510%
30	39000	1416827	8	6	9	0	0.8083%
30	40000	1419820	8	7	16	0	0.9797%
30	41000	1417332	8	7	16	2	0.8014%
30	42000	1420387	7	11	19	77	0.9970%
30	43000	1413738	9	7	11	4	0.5475%
30	44000	1411122	8	7	13	0	0.3875%
30	45000	1414359	9	9	9	0	0.5400%
30	46000	1414829	8	9	24	0	0.5740%
30	47000	1417558	9	7	17	2	0.8165%
30	48000	1415920	7	9	15	108	0.7511%
30	49000	1418597	7	8	3	6	0.8906%
30	50000	1413847	10	7	1	4	0.5577%
30	55000	1417651	10	6	5	110	0.8411%
30	60000	1415214	9	7	13	12	0.6724%

Table 3: Results with 30 orders under alternative budget allocations (continued)

Order size	Γ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
30	65000	1416946	8	7	16	7	0.7731%
30	70000	1417831	9	8	3	111	0.7854%
30	75000	1420858	9	6	6	0	0.9776%
30	80000	1411778	8	7	6	0	0.3454%
30	85000	1414995	6	10	12	6	0.5848%
30	90000	1417374	7	7	42	2	0.7840%
30	95000	1417503	8	10	10	77	0.9999%
30	100000	1413116	8	7	12	5	0.4675%